

LLCTT – Important Operational Considerations (Windows/Linux)

1. General Information

LLCTT (Lamport Logic Clock Teaching Tool) is an educational distributed systems application developed to demonstrate Lamport logical clocks, distributed communication, event ordering, IPv4/IPv6 networking, unicast, broadcast and multicast communication, and logical clock synchronization.

2. Linux Execution

On Linux, LLCTT runs inside a preconfigured containerized environment. Therefore, manual installation of Python APIs or packages is not required because all dependencies are already included inside the container image.

3. Windows Execution Requirements

On Windows, LLCTT does not require manual installation of Python or additional APIs for end users. All required dependencies are already bundled inside the LLCTT executable package generated with Nuitka in standalone mode.

Windows users only need to install the LLCTT executable package and run the application as Administrator.

4. Administrator Privileges

LLCTT must be executed with administrator/root privileges because the application uses multicast sockets, performs low-level networking operations, accesses privileged socket APIs, and modifies or synchronizes the operating system date and time.

5. IPv6 Multicast Considerations

Linux generally provides stable IPv6 multicast support and most multicast scopes operate correctly, including ff03::/16, ff04::/16, and ff05::/16.

Windows is significantly more restrictive regarding IPv6 multicast, especially inside virtualized environments such as VirtualBox and VMware.

6. Recommended IPv6 Multicast Scope on Windows

During testing, the most reliable multicast scope on Windows virtual machines was ff03::/16, particularly ff03::1.

The ff03::/16 scope represents Realm-Local multicast communication, allowing communication across multiple local network segments within the same logical realm.

Examples:

- ff03::1 → Realm-Local All-Nodes
- ff03::2 → Realm-Local All-Routers

7. Important Windows Observation

Although IPv6 multicast using ff03::/16 works on Windows virtual machines, it may still present instability depending on the hypervisor, VM network mode, Windows firewall, NIC driver, and multicast scope handling.

Possible symptoms include packet loss, delayed synchronization, unstable multicast reception, and intermittent communication failures.

8. Recommendations

Linux is strongly recommended for IPv6 multicast experiments.

On Windows, IPv4 Unicast, IPv4 Broadcast, and IPv4 Multicast are generally more stable options. IPv6 multicast on Windows should still be considered experimental depending on the virtualization and networking environment.

9. Recommended Virtual Machine Configuration

For VirtualBox and VMware, use Bridged Networking / Bridged Adapter mode whenever possible. Avoid NAT mode during multicast experiments.

10. Firewall Considerations

Firewalls may block multicast traffic. If communication problems occur, allow LLCTT and UDP traffic through the firewall or temporarily disable the firewall for testing purposes.

11. Final Recommendation

For best operational stability, use Linux whenever possible for IPv6 multicast experiments, execute LLCTT with administrator/root privileges, and prefer Bridged networking in virtualized environments.